Abstract

Understanding the biomechanical response of the cornea following penetrating keratoplasty is crucial for improving surgical outcomes and reducing postoperative complications. This study investigates how graft size, suture configuration, and alignment affect stress distribution, apex displacement, and refractive outcomes using finite element simulations. Sutures were modeled as MPC constraints, wire elements, and 3D solids. Results showed that smaller grafts concentrated stress at the graft-host junction, while larger grafts caused greater displacement. Eccentric placement and uneven suture tension increased stress and instability. Evenly spaced sutures provided better stability. The findings highlight the importance of precise suture placement and graft alignment in PK.